MASTER'S COMPREHENSIVE EXMINATION

Part II **Applied (90 minutes)** **December 3, 2005**

Instructions: Answer all four questions. Open book and open notes.

1. A sample of new automobile models give the following figures for highway mileage in miles per gallon, both highway and city.

Model	Highway	City
A	43	35
B	35	23
C	30	25
D	25	20
E	17	13
F	12	10

Assume that mileage for (highway, city) has a bivariate normal distribution.

(a) Test the hypothesis that the population correlation coefficient is zero vs. nonzero at level .05.

Some calculations:

$16^2 + 8^2 + 3^2 + 10^2 + 15^2 = 658$
$14^2 + 2^2 + 4^2 + 1^2 + 8^2 + 11^2 = 402$
$16 \cdot 14 + 8 \cdot 2 + 3 \cdot 4 + 2 \cdot 1 + 10 \cdot 8 + 15 \cdot 11 = 499$

(b) Suppose now that the sample correlation coefficient calculated in part (a) was based on 52 observations instead of 6. Find a 95% confidence interval for the population correlation coefficient.

2. (a) Using the same data given in problem 1 and using the fact that expected city mileage given highway mileage is linear in highway mileage, find a 95% *prediction* interval for city mileage for a new model car whose highway mileage is 27.

(b) For the regression studied in part (a), what is the percent of total variation of the observed city mileage explained?